# An Exponential Succinctness Gap between Three-Variable Logic and the Calculus of Relations

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STOC 2027 submission　(author video for the PC)

**論文 v8 との差分（用語）：**

- fold → flip
- notice → (newly) distinguish：同じ色の辺で真偽が食い違う。notice では「気づいてどうなるか」（そのゲートで初めて区別が生まれる）が伝わらない
- reference structure → two-witness structure $\mathcal{M}_0$ ／ read・witness edge → $x$-$y$・$x$-$z$ edge
- color-determined（v8）→ color-invariant。bisimulation-invariant などと同じ「同色の点対で値が同じ」を意図

**論文 v8 との差分（内容）：**

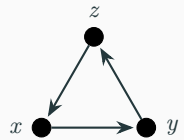
- Key Lemma「$d$ 個を見分ける項は合成 $\geq d/2$」論文でもこの形で先に述べるとよい？
  （v8：各 fold はどこかの合成で notice される ＋ 合成 1 つで高々 2）
- distinguish を定義として明示（v8 は prediction と notice で定義）
- 省略：ループ（$(\hbar, \hbar)$ を含む）、端点 witness（p.11 で "see the paper"）。Cor 1.3（CoR 等式）、Cor 1.8・1.9（固定サイズ・乱択・ランダム入力）、項サイズ $(2 + o(1))k2^k$
- 名前のみ：Thm 1.4（FO$^3$ vs FO$^2$）、Prop 1.5（CoR vs FO$^2$）、Thm 1.6（ARA(3)）、Prop 1.7（RTC）は p.4 で結果の名前だけ
- 追加：導入の FO$^k$ の例（3-cycle、推移性、closed walk）とモデル検査の表

**補足：**

- 投稿版の論文が確定したら定理番号を入れる（v8 の番号）：p.4 の定理 Thm 1.1、各項目 Thm 1.4・Prop 1.5・Thm 1.6・Prop 1.7、Key Lemma → Lemma 3.4・3.5、p.11 の "see the paper" → Lemma 3.5

# $k$-variable logic $\mathrm{FO}^k$

$\mathrm{FO}^k$: first-order logic with $k$ variables.
For example, in $\mathrm{FO}^3$: $\varphi(x,y) := \exists z \left(E(x,y) \wedge E(y,z) \wedge E(z,x)\right)$ (triangle detection)




$\mathrm{FO}^3$ also expresses, e.g.,

- transitivity: $\forall x \forall y \forall z \left(E(x,y) \wedge E(y,z) \to E(x,z)\right)$,
- closed walk of length 4: $\exists x \exists y \left(E(x,y) \wedge \exists z \left(E(y,z) \wedge \exists y \left(E(z,y) \wedge E(y,x)\right)\right)\right)$.

**Model checking**: $\mathcal{M} \models \varphi(a,b)$? ($n$: the number of elements)

- $O(|\varphi| \cdot n^k)$ for $\mathrm{FO}^k$ by table memoization [Var95, PODS]

E.g., $O(n^3)$ for $\varphi$ above: try all $x, y, z$



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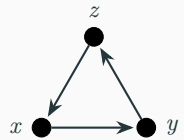
- $\mathrm{FO}^k$ とモデル検査の問題設定
- 素直に計算すると $O(|\varphi| \cdot n^k)$

**口頭補足：**

- 入力は構造 $\mathcal{M}$（要素数 $n$）、式 $\varphi$、要素 $a, b$

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We consider binary relation structures with $n$ elements:

| syntax | model checking | |
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| $\mathrm{FO}^3$ | $2^{O(\lvert\varphi\rvert)} \cdot n^\omega$<br>no $2^{o(\lvert\varphi\rvert)} \cdot n^{\omega-\varepsilon}$<br>under 3-OV Hypothesis | [NU27, PODS] [Wil14, CSL-LICS, for PNF]<br>[NU27, PODS] |
| $\mathrm{CoR}$ [Tar41, JSL]<br>Tarski's Calculus of Relations | $O(\lvert\varphi\rvert \cdot n^\omega)$ | |

Boolean matrix algorithm of $\mathrm{FO}^3$ model checking via $\mathrm{FO}^3 \rightsquigarrow \mathrm{CoR}$ [Tar41, JSL; TG87]:

$$\exists z.\ P(x,z) \wedge Q(z,y) \quad \rightsquigarrow \quad (P;Q)(x,y)$$

composition = Boolean matrix multiplication

To obtain the form of $\exists z.\ _ \wedge _$, we use DNF translation.
The translation uses $2^{O(\lvert\varphi\rvert)}$ compositions [Nak22, IEAMF].



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DNF translation is used, for example:

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**Problem (Open [FGL+15, Inf. Sci.; Nak22, JLAMP; NU27, PODS])**

*Can we prove that there is no polynomial-size translation from* $\mathrm{FO}^3$ *to* $\mathrm{CoR}$*?*

Partial answer [Nak22, JLAMP]: yes for **existential** $\mathrm{CoR}$ [Nak23, LICS] ($\equiv$ existential $\mathrm{FO}^3$), via preservation under **extensions**, which fails for full $\mathrm{CoR}$.



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# Outline

1. Our results

2. Technical overview

3. Conclusion

## Our results

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**Thm. (This paper)**

*There is no polynomial-size translation from* $\mathrm{FO}^3$ *to* $\mathrm{CoR}$*.*
More precisely, there are $\mathrm{FO}^3$ formulas $\varphi_1, \varphi_2, \ldots$ with $|\varphi_k| = O(k)$ such that every $\mathrm{CoR}$ term equivalent to $\varphi_k$ needs $\geq 2^{k-1}$ compositions, even with shared subterms.

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Our proof technique applies to all of them!



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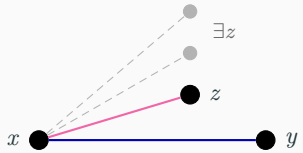


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# The hard formula $G_k$

$$G_k(x,y) \coloneqq \exists z \bigwedge_{i=1}^{k} \left[ (P_i(x,y) \wedge Q_i(x,z)) \vee (Q_i(x,y) \wedge P_i(x,z)) \right] \qquad (|G_k| = O(k))$$




$G_k(x,y)$ holds iff there is $z$ such that
for every $i \in [k]$,
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CoR: $2^k$ compositions

$$G_k \equiv \bigcup_{A \subseteq [k]} \left[ \left( \bigcap_{i \in A} P_i \cap \bigcap_{i \notin A} Q_i \right) \cap \left( \left( \bigcap_{i \in A} Q_i \cap \bigcap_{i \notin A} P_i \right) ; \mathbf{1} \right) \right]$$



**ページ概要：**

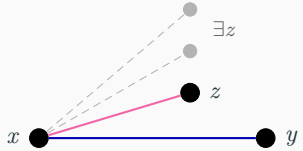
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**口頭補足：**

- 前の例と同じく DNF で展開すると合成 $2^k$ 個

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# Outline

1. Our results

2. Technical overview

3. Conclusion

# Proof outline (high level)

**Goal Lemma** Every $\mathrm{CoR}$ term equivalent to $G_k$ needs $2^{\Omega(k)}$ compositions.

**Approach** We construct structures $\mathcal{M}_0$ and $\mathcal{M}_1, \ldots, \mathcal{M}_{2^k}$ such that

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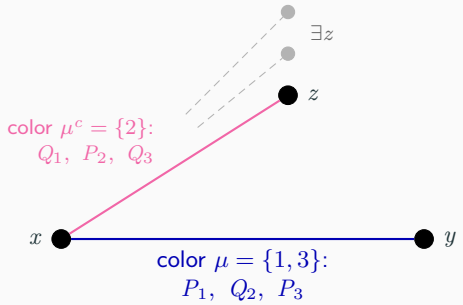


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## Structures as colored complete graphs

**View** structures as **complete graphs**, colored with $2^k$ colors $\mu \subseteq [k]$: $P_i$ iff $i \in \mu$. $Q_i$ is used as the complement of $P_i$ (i.e., $i \notin \mu$).




for every $i \in [3]$,
the $x$-$y$ and $x$-$z$ edges
have labels $P_i$ and $Q_i$, in either order

☛ $G_k(x, y)$ holds iff some $x$-$z$ edge has the color $\mu^c$, where $\mu$ is the color of the $x$-$y$ edge.



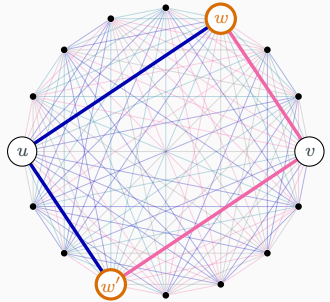
**ページ概要：**

- 構造を色付き完全グラフとして見る
- $G_k(x, y)$ は $x$-$y$ の色 $\mu$ に対して、$x$-$z$ の色が $\mu^c$ であるような $z$ が存在することと同値

**口頭補足：**

- 各辺で $P_i$ と $Q_i$ のちょうど一方が成り立つ

# The two-witness structure $\mathcal{M}_0$




$\mathcal{M}_0$: $O(k4^k)$ vertices, constructed by, e.g., a random coloring [JMT95, Algebra Univers.]

**Two-witness property** For all $u \neq v$ and all colors $\boldsymbol{\mu}, \boldsymbol{\nu}$, there are at least two **witnesses** $w$ such that the $u$-$w$ and $w$-$v$ edges have colors $\boldsymbol{\mu}$ and $\boldsymbol{\nu}$, respectively.

**Every CoR term $t$ is color-invariant on $\mathcal{M}_0$**: $\mathcal{M}_0 \models t(u, v) \iff \mathcal{M}_0 \models t(u', v')$ whenever the $u$-$v$ and $u'$-$v'$ edges have the same color.

Proof by induction on $t$: the case $t = t_1 \,;\, t_2$ follows from the (two-)witness property.

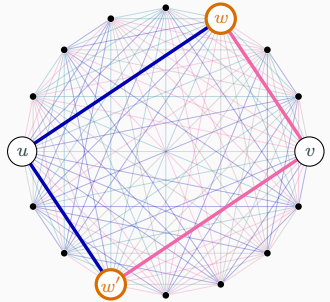
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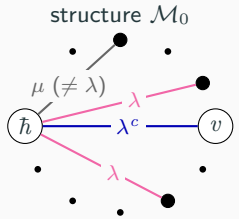


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# $\mathcal{M}_\lambda$: flipping at the hub $\hbar$

**Hub** $\hbar$: a fixed arbitrary vertex of $\mathcal{M}_0$.




$\hbar$ has $\lambda$-edges

For $v$ with the $\hbar$-$v$ edge of color $\lambda^c$: $\mathcal{M}_0 \models G_k(\hbar, v)$ but $\mathcal{M}_\lambda \not\models G_k(\hbar, v)$.
Flipping each of the $2^k$ colors $\lambda$ gives $\mathcal{M}_1, \ldots, \mathcal{M}_{2^k}$.
**Definition** $r$ **distinguishes** $\mathcal{M}_\lambda$ from $\mathcal{M}_0$ if $\mathcal{M}_0 \models r(u, v)$ but $\mathcal{M}_\lambda \not\models r(u', v')$ (or vice versa) for some $u$-$v$ edge in $\mathcal{M}_0$ and $u'$-$v'$ edge in $\mathcal{M}_\lambda$ of the same color.
$\rightsquigarrow$ Each composition newly distinguishes at most two of them (next slides).



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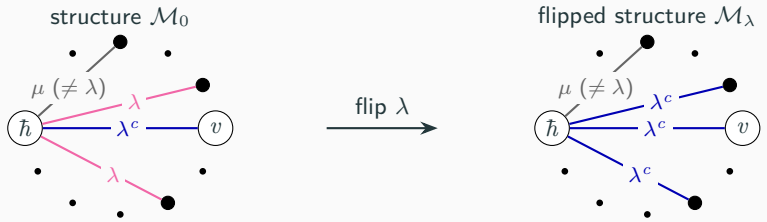
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- 各合成が新たに区別するのは高々 2 個（次の 2 ページ）

**口頭補足**：

- 補・逆・和・積は点対ごとの演算なので、新たに区別することはない（口頭）

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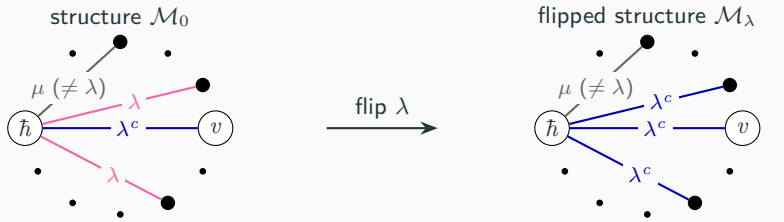
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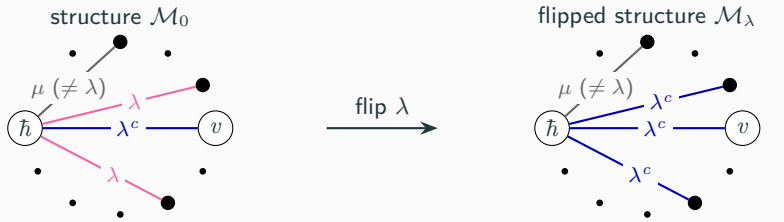
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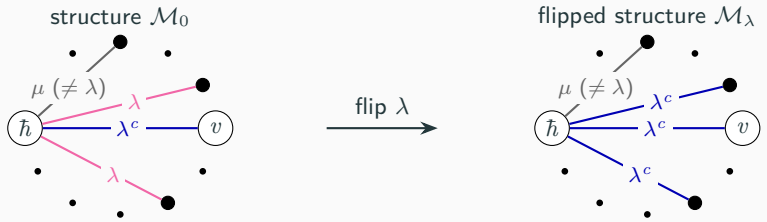
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**Q.** If $t$ and $s$ do not distinguish $\mathcal{M}_\lambda$ from $\mathcal{M}_0$, when can $t\,;s$ newly distinguish $\mathcal{M}_\lambda$?

**Case $\hbar \notin \{u, v\}$:** Never: $t\,;s$ cannot distinguish $\mathcal{M}_\lambda$ from $\mathcal{M}_0$ at such $(u, v)$.

Similarly for the converse direction (from $\mathcal{M}_\lambda$ to $\mathcal{M}_0$).



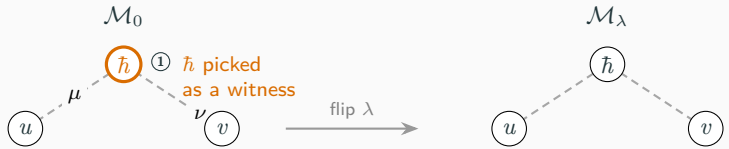
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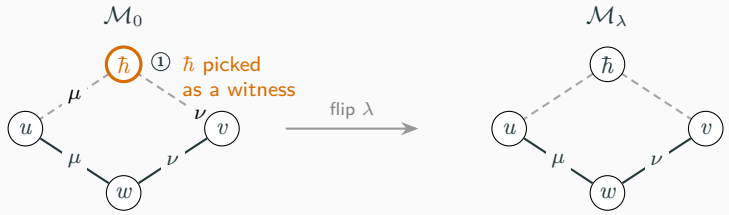
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② The **two-witness property** shows there is $w$ with the same colors $(\mu, \nu)$

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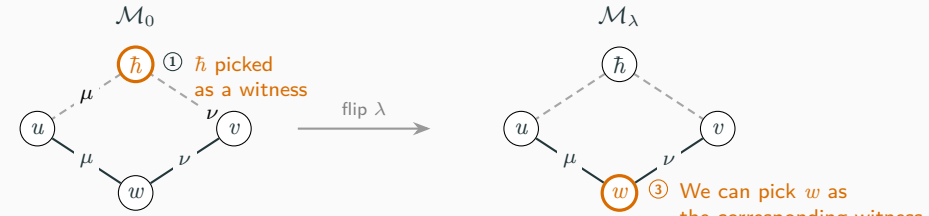
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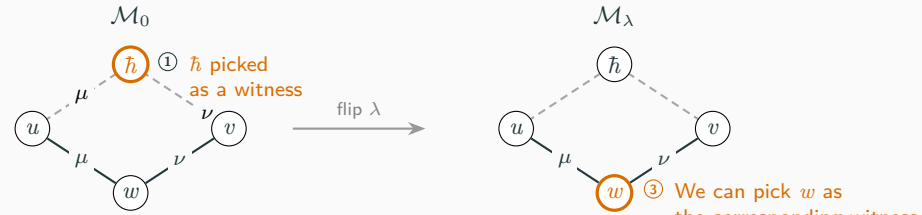
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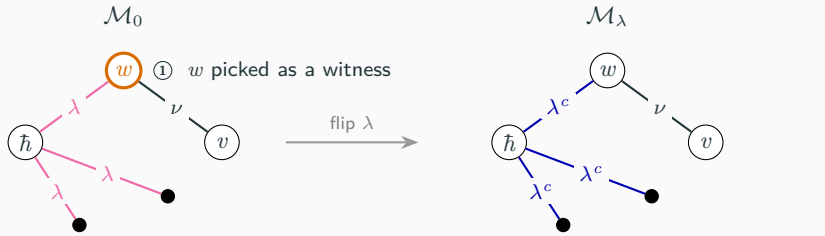


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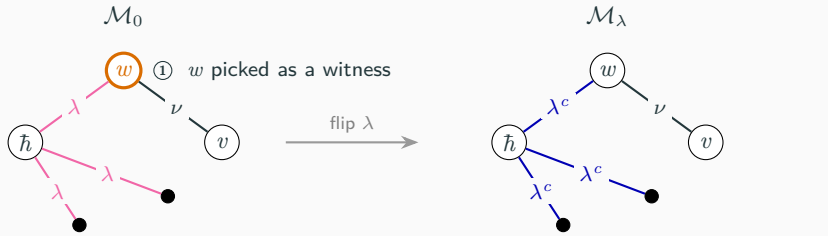


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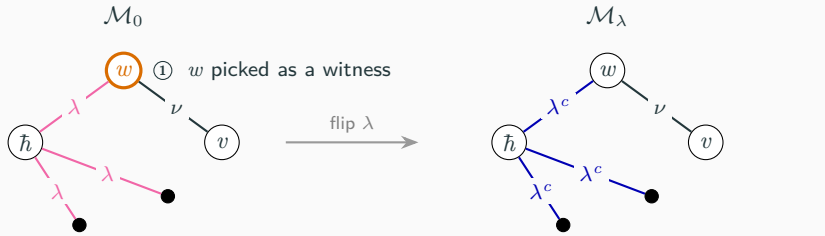


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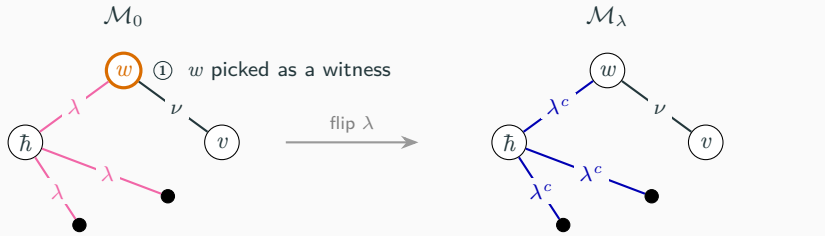


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# Outline

1. Our results

2. Technical overview

3. Conclusion

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*There are* $\mathrm{FO}^3$ *formulas* $\varphi_1, \varphi_2, \ldots$ *with* $|\varphi_k| = O(k)$ *such that every* $\mathrm{CoR}$ *term equivalent to* $\varphi_k$ *needs* $\geq 2^{k-1}$ *compositions.*

Our proof also applies to $\mathrm{FO}^3$ vs. $\mathrm{FO}^2$, matrix query languages, and $\mathrm{CoR}$ with RTC.

**Key Idea** the two-witness structure $\mathcal{M}_0$ with its $2^k$ flipped structures.
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**Problem (Open Questions)**

- *a superpolynomial gap over a* ***fixed*** *finite signature? (e.g., one binary relation)*
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**Thank you for your attention!**



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- 結果：合成 $\geq 2^{k-1}$
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